

Combinatorial strategies employing nutraceuticals for cancer development.



Cancer is the second leading cause of death worldwide. Therefore, the fight against cancer is one of the most important areas of research in medicine, and one that possibly contributes to the increased interest in chemoprevention as an alternative approach to the control of cancer. Cancer prevention by nutraceuticals present in fruits and vegetables has received considerable attention because of their low cost and wide safety margin. A substantial amount of evidence from human, animal, and cell culture studies has shown cancer chemopreventive effects from these natural products. However, single-agent intervention has failed to produce the expected outcome in clinical trials; therefore, combinations of nutraceuticals are gaining increasing popularity. Thus, combinations of nutraceuticals that mimic real-life situations and are competent in targeting multiple targets with very little or virtually no toxicity are needed. In this review, we summarize the results of those studies that report combinatorial cancer chemopreventive action of various nutraceuticals and their combinations with anticancer drugs. © 2011 New York Academy of Sciences.